Velocity graphs and components

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Learning target: Read a velocity graph and describe motion in two directions.

Suggested first pass: 1, 2, 3, 9. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 1; 3 course days later.*
52 m if the 6 m/s rightward motion continues. Check a later timed position against the prediction.

02. Explain a worked example (worked)
Rectangle height 3 m/s, width 2 s; area 6 m east.

03. Steady does not mean stopped (model)
Steady motion at +3 m/s; displacement +12 m.

04. Move, then wait (practice)
6 m west; waiting adds zero displacement and does not undo the trip.

05. One diagonal motion (reason)
4 m/s east and 3 m/s north. Both components occur simultaneously; the actual motion is diagonal.

06. Make the components visible (reason)
Diagonal from (0, 0) to (3, 2); components 3 m/s east and 2 m/s north.

07. Fix the speed claim (reason)
Components are perpendicular, not along one line. The diagonal is 5 m/s as supplied; 4+3 is not its magnitude. No square-root calculation required.

08. Choose a method (mixed) *Review: Unit 1, Day 3; 1 course days later.*
v=(5-15)/3=-3.333 m/s, left. Straight descending graph between the stated endpoints.

09. Exit check (exit)
Displacement -10 m, or 10 m in the negative direction. Second sketch points southeast, with east component twice the south component.

10. STRETCH: Apply it another way (stretch)
Example +2 m/s for 2 s then -2 m/s for 2 s: areas cancel, distance 8 m.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 1; 3 course days later.*
v=-3 m/s; x(8)=-6 m. Assumes constant velocity; compare another timed position with its uncertainty.

02. Explain a worked example (worked)
Rectangle height 3 m/s, width 2 s; area 6 m east.

03. Add signed areas (model)
First 4 s: +12 m. Last 2 s: -4 m. Displacement +8 m; distance 16 m. Motion positive then negative.

04. Resolve a northeast velocity (practice)
v_x=14 cos 35°=11.5 m/s; v_y=14 sin 35°=8.03 m/s. East component larger because the direction is closer to east.

05. Change the reference axis (reason)
v_x=-10 sin 20°=-3.42 m/s; v_y=10 cos 20°=9.40 m/s. North component is adjacent to the stated angle.

06. Draw velocity and displacement (reason)
Speed 5 m/s. Displacement (8,-6) m, magnitude 10 m, same southeast direction.

07. Two horizontal graphs (reason)
x=5 m: stationary at that position, v=0. v=5 m/s: constant positive motion with zero acceleration; position changes 5 m each second.

08. Choose a method (mixed) *Review: Unit 1, Day 3; 1 course days later.*
v=5 m/s; position increases 5 m each second. x(7)=27 m.

09. Exit check (exit)
Displacement 6-2=4 m; distance 6+2=8 m. Need reference axis and direction/sense of angle.

10. STRETCH: Apply it another way (stretch)
Positive displacement 12 m; return takes 6 s. Distance 24 m in 9 s gives 8/3 m/s; average velocity zero.